SP19_340-001_Exam1_D (1)

Saturday, October 1, 2022 12:21 PM

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Math 340-004: Elementary Matrix and Linear Algebra

MIDTERM ONE

February 13, 2019

TABLE OF DISCUSSION SECTIONS

Section	Meets		Leader
361	M	8:50 - 9:40 am	Huang
362	M	9:55 - 10:45 am	Huang
363	M	11:00-11:50 am	Mohamed
364	M	12:05-12:55 pm	Mohamed
365	M	1:20-2:10 pm	Mohamed

Name: ________________ Section#: ______ NetID: __________

Key

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: ____________________________

READ THE FOLLOWING INFORMATION.

- This is a **60-minute** exam. It consists of 6 problems on 7 pages. It is your responsibility to make sure that you have a complete exam.
- You may use a standard scientific calculator, but not one which has sizeable stored memory.
- Books, notes, and other aids are not allowed.
- You may use the backs of the pages or any additional pages for scratch work, but it will not be graded. ***Do not unstaple or remove pages as they can be lost in the grading process.***
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.

DO NOT BEGIN THIS EXAM UNTIL SIGNALED TO DO SO.

1. (9 points) Let

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 4 \\ 3 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 3 \\ -1 & -1 \\ -5 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 \\ 4 & -8 \end{bmatrix}.$$

Compute the following (if possible). If the computation is not possible, explain why.

(a) $2A - B$.

$$= \begin{bmatrix} 2 & 4 \\ -2 & 8 \\ 6 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 3 \\ -1 & -1 \\ -5 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & 9 \\ 11 & -2 \end{bmatrix}$$

(b) $A^T B - C$.

$$= \underbrace{\begin{bmatrix} 1 & -1 & 3 \\ 2 & 4 & 1 \end{bmatrix}}_{2\times 3 \checkmark} \underbrace{\begin{bmatrix} 1 & 3 \\ -1 & -1 \\ -5 & 4 \end{bmatrix}}_{3\times 2 \checkmark} - \begin{bmatrix} 1 & 2 \\ 4 & -8 \end{bmatrix}$$

$$= \begin{bmatrix} -13 & 16 \\ -7 & 6 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 4 & -8 \end{bmatrix} = \begin{bmatrix} -14 & 14 \\ -11 & 14 \end{bmatrix}$$

(c) $BA^T + C$.

$3\times 2 \;\; 2\times 3 \qquad 2\times 2$

$3\times 3 + 2\times 2 \rightarrow$ X can't do it. dimensions don't match.

2. (12 points) For each statement given, determine if it is true or false. In either case, offer a short explanation why. You should assume that the sizes of all matrices and vectors given are appropriate for the given operations.

(a) A system of equations can have exactly two solutions.

False, if it has 2 it has infinitely many.

(b) If A is an invertible matrix then $Ax = 0$ has an infinite number of solutions.

False, it only has the trivial soln $\vec{x} = \vec{0}$.

(c) Matrix multiplication is commutative.

False, AB may not equal BA.

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\neq$$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

(d) Let A and B be two $n \times n$ matrices. Then $(AB)^T = A^T B^T$.

False, $(AB)^T = B^T A^T$.

(e) Let A and B be invertible $n \times n$ matrices. Then the product AB is invertible.

True! $(AB)^{-1} = \underbrace{B^{-1}A^{-1}}_{\text{both defined.}}$

(f) If A is an $n \times n$ matrix then $A + A^T$ is symmetric matrix.

True b/c $(A + A^T)^T = A^T + (A^T)^T$

$= A^T + A = A + A^T$

3. (12 points) The following questions ask you to give definitions. You should aim to give the specific definitions used in this course.

(a) What is a homogeneous linear system?

Were every equation is set equal to 0, ie.

$$A\vec{x}=\vec{0}.$$

(b) What is a singular matrix?

It is noninvertible.

(c) What does it mean to say that two systems of equations are equivalent?

They have the same solutions.

(d) What does it mean to say that matrix multiplication is associative?

$$(AB)C = A(BC)$$

(e) What is the matrix transformation associated to a matrix $A \in \mathbb{R}^{n\times m}$?

$$\vec{x} \to A\vec{x}$$

(f) What is a diagonal matrix?

only has entries on the main diagonal:

$$A = \begin{bmatrix} a_{11} & & 0 \\ & a_{22} & \\ 0 & & \ddots \; a_{nn} \end{bmatrix}$$

4. (9 points) Let

$$A = \begin{bmatrix} 3 & a \\ 2 & 1 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

(a) Find all values of a so that A is singular.

$\det(A) = 0$:

$3 - 2a = 0$

$\boxed{a = 3/2}$

(b) If $a = 0$ solve the linear system $B\vec{x} = \vec{b}$ if $B^{-1} = A$.

$\vec{x} = B^{-1}\vec{b} = A\vec{b} = \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

$= \boxed{\begin{bmatrix} -3 \\ 0 \end{bmatrix}}$

(c) If $a = 2$ express $Ax = b$ as a system of linear equations.

$3x + 2y = -1$

$2x + y = 2$

5. (9 points) Consider the linear system of equations

$$\begin{aligned} x + y - 2z &= 5 \\ 2x + 3y + 4z &= 2. \end{aligned}$$

(a) Determine a matrix A and vector $\vec{b}$ so that the given system is equivalent to $A\vec{x} = \vec{b}$.

$$A = \begin{bmatrix} 1 & 1 & -2 \\ 2 & 3 & 4 \end{bmatrix} \qquad \vec{b} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

(b) Which of the following best describes the solution set to the given linear system. Explain your choice using only geometric reasoning.

i. The solution set is empty.
ii. The solution set is trivial.
iii. The solution set is a single point in $\mathbb{R}^3$
iv. The solution set is a line in $\mathbb{R}^3$
v. The solution set is a plane in $\mathbb{R}^3$.

(c) Completely determine the solution set to the system.

z free variable.

$$\left[\begin{array}{ccc|c} 1 & 1 & -2 & 5 \\ 2 & 3 & 4 & 2 \end{array}\right] \xrightarrow{-2r_1+r_2 \to r_2} \left[\begin{array}{ccc|c} 1 & 1 & -2 & 5 \\ 0 & 1 & 8 & -8 \end{array}\right]$$

$z = t$

$y = -8 - 8t$

$x = 5 - y + 2z$

$= 5 + 8 + 8t + 2t$

$= 13 + 10t$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 13 + 10t \\ -8 - 8t \\ t \end{bmatrix}$$

6. (9 points) You wish to find an equation whose graph is a parabola which passes through the points $(-1,2)$, $(1,0)$, and $(2,2)$. The general form for such an equation is the quadratic $y = Ax^2 + Bx + C$, where A, B, C are some real numbers. Thus, if you could determine the values of A, B, and C, then you would have such an equation.

(a) Explain why it must be true that

$$\begin{aligned} A+B+C &= 0 = y(1) = A\cdot 1^2 + B\cdot 1 + C \\ A-B+C &= 2 = y(-1) = A(-1)^2 + B(-1) + C \\ 4A+2B+C &= 2 = y(2) = A\cdot 2^2 + B\cdot 2 + C \end{aligned}$$

(b) Solve the above linear system using an elimination method and give the quadratic equation.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & 2 \\ 4 & 2 & 1 & 2 \end{array}\right] \xrightarrow[-4r_1+r_3 \to r_3]{-r_1+r_2 \to r_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -2 & 0 & 2 \\ 0 & -2 & -3 & 2 \end{array}\right] \xrightarrow{-\frac{1}{2}r_2 \to r_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & -2 & -3 & 2 \end{array}\right]$$

$$\xrightarrow{2r_2+r_3 \to r_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & -3 & 0 \end{array}\right]$$

$\to A+B+C=0 \to A=-B=1.$

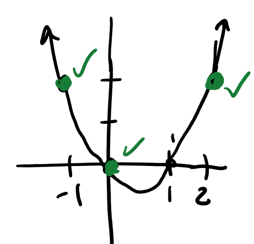
$\to B=-1$

$\to -3C=0 \to C=0$

Answer:

$y = x^2 - x$

$\to$ check:



(c) Suppose instead that you wanted to solve the above problem but only given two points: $(-1,2)$ and $(1,0)$. Explain why there are an infinite number of such equations.

↳ only 2 equations + 3 unknowns.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & 2 \end{array}\right] \xrightarrow{-r_1+r_2 \to r_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -2 & 0 & 2 \end{array}\right] \xrightarrow{-\frac{1}{2}r_2 \to r_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & -1 \end{array}\right]$$

C is a free variable.